

Net change



1

a 40 kangaroos

b 8 months

2

a i No ii Yes iii No iv Yes

b 3000

c i Yes ii Yes iii No iv No

d 36 000

3

a $-2.02^{\circ}\text{C}/\text{minute}$ b -41.74°C

c $-4.64^{\circ}\text{C}/\text{minute}$

4

a $-\frac{24}{\pi}$

b $-\frac{30}{\pi}$

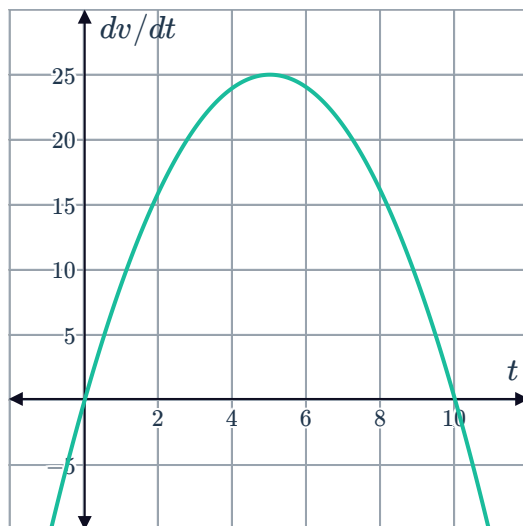
5

a 5818.82 L

b 175.71 L

6

a



b 25 L/min

c $V = 5t^2 - \frac{t^3}{3}$

d 15 minutes

7

a 3700 tonnes/year b 136 tonnes/year c 100 tonnes/year d 4140 tonnes

8

a $M = \frac{81t^2}{2} - \frac{t^4}{4} + 1000$

b 2568 kg

c $t = 9$



9

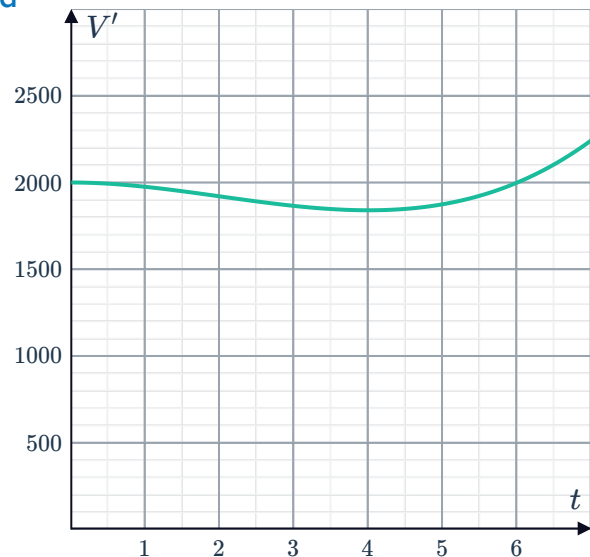
a 2000 L/h

b

t	-1	0	1	4	6
V''	75	0	-45	0	180

c $t = 7$ at 2245 L/h

d



e 5831.25 units²

f The total change in the amount of water in the tank in the first 3 hours is 5831.25 L.

10

a $\frac{dV}{dt} = -25$

b $V = 15\,625 - 25t$

c $t = 625$

11

a $m = \frac{2t^{\frac{3}{2}}}{360} + \frac{t^3}{4320}$

b $\frac{176}{135}$ g

c 0.004 g

12

a 145 customers

b 184 customers

c 134 customers

Cost, revenue and profit applications

13

a \$6222

b \$1037/item



14

- a $P(x) = 0.0004x^3 + 0.6x^2 - 4.1x - 60$ b \$2730.20
c \$9493.40

15

- a b \$40 000
i 40 000 units² ii 32 500 units²
c \$34 500 d \$5500
e -\$5000

16

- a $C = 0.1x^4 + 2x^2 + 2000$ b \$632 000

17

- a $C(x) = 3x^3 - 35x^2 + 500x + \text{initial costs}$
b \$532 000

18

- a $C(x) = 240x^{\frac{1}{2}} + \text{initial costs}$ b \$960

19

- a -\$17 601.30 b -\$704 700
c -\$23 490/item

20

- a $P(x) = 3x^2 + 2x^3 + 5x$ b \$56 850

21

- a \$5.03 b \$4.57

22

- a $R = 401t - \frac{250}{(t+1)^2} + 250$ b \$449 611.11/week
c \$402 842.40

23

- a i \$81.62 ii \$9000 iii \$90
b i \$31.42 ii \$1607.70 iii \$16.08

